

ISHI VARSHNEY

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EDUCATION

University of North Carolina at Chapel Hill

Expected May 2028

B.S. Computer Science & B.S. Biostatistics (Gillings School of Global Public Health) | GPA: 3.86

Awards & Honors: Garmin Next-Gen Scholarship (\$5,000), Honors Carolina (top 10%), National Merit Finalist, Dean's List (all semesters)

Coursework: Data Structures/Analysis, Systems Fundamentals, Computer Organization, Foundations of Software Engineering, Models of Languages & Computation, Discrete Mathematics, Linear Algebra, Intro to Probability

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, R, SQL/PostgreSQL | **Frameworks & Tools:** AWS, Docker, Kubernetes, Git/GitHub, Svelte, Angular, Tailwind CSS, FastAPI, PostgreSQL, Postman, Tableau, Figma, Matplotlib | **Systems & Hardware:** Arduino, Raspberry Pi, Linux | **Skills:** Full-Stack Development, REST APIs, Data Analysis, Robotics, Machine Learning, Three-tier Architecture, HTTP Protocols

EXPERIENCE

Laboratory Assistant — PAWS Guide Dog Project | Experimental Engineering Lab, UNC CS Department

Dec 2025 – Present

- Developing PAWS (Personal Autonomous Walking Support), an autonomous robotic guide system for visually impaired users using the Boston Dynamics SPOT platform, integrating voice-command navigation, waypoint-based routing, and real-time obstacle avoidance.
- Designed and implemented hardware-software interfaces using Arduino (C/C++) and Raspberry Pi, creating a specialized guidance handle with tactile feedback and emergency stop functionality for safe human-robot interaction.
- Built navigation and command-processing workflows that translate natural-language destination requests into predefined route execution using the SPOT Python SDK, GraphNav, and Google Maps integrations.

Software Engineering Intern | Fletch, Cary, NC

May 2025 - Aug 2025

- Developed an AI-powered customer journey optimization tool that tracked user movement across AI chat agents, quote flows, and policy binding experiences to identify conversion drop-off points and improve customer experience workflows.
- Built a Figma-to-JSON conversion tool that automated translation of UI designs into Fletch's proprietary configuration format, streamlining development workflows for customer-facing insurance products.
- Built and tested REST API integrations with Postman and JSON workflows for insurance quote retrieval, analytics tracking, and payment systems while collaborating with engineers to debug, test, and deploy platform features.

Undergraduate Teaching Assistant — COMP 110 | UNC Chapel Hill

Dec 2025 – Present

- Lead review sessions and grading support for 900+ students, reinforcing Python fundamentals, debugging techniques, and algorithmic problem-solving.
- Developed a full-stack scheduling web application (JavaScript/TypeScript) for a 60+ person TA team, eliminating manual scheduling conflicts and improving course logistics.

PROJECTS

LearnWithAI — AI-Powered Course Management Platform | UNC COMP 423

April 2026

- Developed an AI-powered learning platform with a team using Angular, FastAPI, PostgreSQL, and Azure OpenAI APIs to analyze exams and generate personalized study materials.
- Implemented asynchronous workers with Dramatiq/RabbitMQ and real-time WebSocket updates for AI processing tasks.
- Built containerized microservices with REST APIs, shared Python service layers, testing pipelines, and Kubernetes deployment.

LunaCycle — Full-Stack Menstrual Health Platform | HackNC Hackathon

Oct 2025

- Built a full-stack menstrual health platform using Java, HTML/CSS, and the Gemini API, enabling cycle tracking, diary logging, and AI-powered health recommendations personalized to user history.
- Reduced Gemini API response latency by 40% through optimization of API request workflows; led end-to-end development and UI architecture for a 4-person team using Figma and agile collaboration practices.

LEADERSHIP & INVOLVEMENT

Outreach Director | HackNC

Apr 2026 – Present

- Drive sponsor acquisition, university/community partnerships, and hacker recruitment for HackNC, targeting 500+ participant attendance; manage outreach pipeline across 10+ schools and 15+ corporate sponsors.

TA & Curriculum Lead | Girls Who Code

Sep 2025 – Present

- Instruct 20+ middle and high school students in Web Development (HTML/CSS/JavaScript) and Data Visualization with Matplotlib/Python; developed 3 reusable lesson modules adopted across multiple sessions.